

Arctic Network

CPU TIME LIMIT

2 seconds

MEMORY LIMIT

1024 MB

DIFFICULTY

medium (2.9)

The Department of National Defence (DND) wishes to connect several northern outposts by a wireless network. Two different communication technologies are to be used in establishing the network: every outpost will have a radio transceiver and some outposts will in addition have a satellite channel.

Any two outposts with a satellite channel can communicate via the satellite, regardless of their location. Otherwise, two outposts can communicate by radio only if the distance between them does not exceed D , which depends of the power of the transceivers. Higher power yields higher D but costs more. Due to purchasing and maintenance considerations, the transceivers at the outposts must be identical; that is, the value of D is the same for every pair of outposts.



Your job is to determine the minimum D required for the transceivers. There must be at least one communication path (direct or indirect) between every pair of outposts.

Input

The first line of input contains N , the number of test cases. The first line of each test case contains $1 \leq S \leq 100$, the number of satellite channels, and $S < P \leq 500$, the number of outposts. P lines follow, giving the (x, y) coordinates of each outpost in km (coordinates are integers between 0 and 10 000).



Output

For each case, output should consist of a single line giving the minimum D required to connect the network. Output should be specified to 2 decimal points.

Sample Input 1

```
1
2 4
0 100
0 300
0 600
150 750
```

Sample Output 1

```
212.13
```